

Valuing Interest Rate Swaps

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1 Introduction

This is the documentation behind the pricing of interest swaps and Forward Rate Agreements. This is meant to explain the mathematics behind the code which does the mathematical heavy lifting to give the prices of these derivatives. I have taught myself these concepts using "An Introduction to Quantitative Finance" by Stephen Blyth as well as Howard Cobb's book on Interest Rate Derivatives.

2 Interest Rate Swaps

Firstly, let's define Interest Rate Swaps. Interest Rate Swaps are an exchange of a series of cash flows at agreed upon dates between two different parties. In this case the cash flows that are exchanged must be in the same currency.

Possible motivations to enter an interest rate swap can be hedging or speculation. Hedging can involve a few things: a firm possibly trying to manage their duration, interest rate sensitivity, or their balance sheet as if they might have assets and liabilities with different interest rate sensitivity. So a swap allows them to match the duration of their assets and liabilities, if their liability duration is larger than their asset duration. As they synthetically receive the duration experience of a bond, as we will show later, a swap can be written as the difference between bonds. Firms can also speculate on interest rates if they believe rates will rise they will pay the fixed and receive the floating and vice versa.

Now we can start to address pricing these swaps. We can start by defining a zero coupon bond. This is an instrument that is priced at a discounted face value and the face value is paid back at maturity. So the return is based on the rate r , that the face value is discounted at. So we can write for a zero coupon bond paying a face value of 1 at time T its maturity. Its current price at time t is:

$$Z(t, T) = \frac{1}{\left(1 + r_i^{(T-t)}\right)}$$

Above we can see we are dividing our face value, 1, by some discount factor. This discount factor is made from the risk free rate, the rate at which an investor can receive a return with taking on zero risk. We define our risk free rate from current time t to different maturities T_i as being r_i . This allows us to discount the face value of a zero coupon bond for different expiry dates. The question I will answer next is, instead of looking at the risk free rate from time t to T_2 what is the risk free rate from T_1 to T_2 . This is what we call the forward rate, f_{12} :

$$(1 + r_1)^{(T_1-t)}(1 + f_{12})^{(T_2-T_1)} = (1 + r_2)^{(T_2-t)}$$

Solving for f_{12} , we gain:

$$f_{12} = \frac{r_2(T_2 - t) - r_1(T_1 - t)}{T_2 - T_1}$$

This is the Forward rate between the period T_1 to T_2 , where $T_1 < T_2$. This is the rate someone can borrow or lend from those two periods in time.

Interest rate derivatives depending on the nation payout mostly on a risk free rate, for the UK it is SONIA and the US SOFR. For ease and notion we will call $S_T[t, T + \alpha]$. Where α is the SOFR or SONIA rate we are using i.e 12 month SONIA rate would be $S_T[t, T + 1]$. This can stand for either SONIA or SOFR.

We have established how to calculate a forward rate, and defined the different risk free rates used in the UK and US. Now we can start to value our first interest rate derivative, the Forward Rate Agreement. This is a forward contract to exchange cash flows at maturity date $T + \alpha$. The buyer pays a fixed rate K , and receives a rate $S_T[T, T + \alpha]$. Therefore the payout of the FRA at time $T + \alpha$ is $\alpha(S_T[T, T + \alpha] - K)$. No cash flows are exchanged until the maturity date. This will be a useful idea to build on when instead of looking at cash flows being exchanged at time $T + \alpha$ we exchange cash flows at time T , this is an arrears FRA.

We can now write our forward rate of $S_T[T, T + \alpha]$ in terms of ZCB's. We can show this via replication, as Blyth does in his book "An Introduction to Quantitative Finance", we can buy one ZCB maturing at time T and going short $(1 + \alpha K)$ ZCB's maturing at time $T + \alpha$. At the expiration date T , we place the face value of the ZCB, 1 into an account paying $S_T[T, T + \alpha]$. This means at the maturity date of the short ZCB's the total position value is:

$$(1 + \alpha S_T[T, T + \alpha]) - (1 + \alpha K)$$

This makes sense as we have to pay back the short position and we receive the interest from T to $T + \alpha$ on 1 pound. We can also notice this is the payout of the FRA, therefore it is the value of the FRA at time t . so we can say our value of the portfolio of assets we hold at time t is

$$V_K(t, T) = Z(t, T) - (1 + \alpha K)Z(t, T + \alpha)$$

The value of K which the value of our forward contract has a value zero at time t , is our forward rate. Following from this:

$$\alpha S_t[T, T + \alpha] Z(t, T + \alpha) = Z(t, T) - Z(t, T + \alpha)$$

So solving for our forward rate:

$$S_t[T, T + \alpha] = \frac{Z(t, T) - Z(t, T + \alpha)}{\alpha Z(t, T + \alpha)}$$

So we have our forward and discount rate. Now we can tackle valuing the two sides of the swap. As previously defined, our swaps consist of a fixed and floating rate leg depending on if you are buying or selling the swap you receive either one of these legs. Lets take the case of you are a buyer of an interest rate swap with maturity at time T , starting at time T_0 . We can write our interest rate swap as:

$$V_k^{SW}(t) = V_K^{FL}(t) - V_K^{FXD}(t)$$

Where $V_k^{SW}(t)$ is the value of the swap at time t , and our terms are the value of the floating leg and the fixed leg respectively. Now we can value each leg, let's start by valuing our fixed leg. Our fixed leg is a just a series of fixed cash flows, αK until the maturity date, at dates T_i . so we can say the value of the fixed leg is

$$V_K^{FXD}(t) = \sum_{i=1}^n (\alpha K) Z(t, T_i) = K \sum_{i=1}^n \alpha Z(t, T_i)$$

As Blyth does in his book, we will also define our pv01 as being:

$$P_t[T, T + \alpha] = \sum_{i=1}^n \alpha Z(t, T_i)$$

This is the present value of receiving 1 pound at every date T_i . Now we can find the value of our floating leg. At every date T_i , we receive $\alpha S_t[T, T + \alpha]$. So our value is just the sum of these cash flows present value. This gives us:

$$V_K^{FL}(t) = \sum_{i=1}^n \alpha S_t[T, T + \alpha] Z(t, T_i)$$

Which as we have shown from before can write as being

$$V_K^{FL}(t) = \sum_{i=1}^n Z(t, T_{i-1}) - Z(t, T_i) = Z(t, T_0) - Z(t, T_n)$$

This makes sense as the return of the floating rate is just receiving a pound at the start of the swap and then having to pay it back at time T_n . So from this now we have all the pieces to value our swaps. We have set an arbitrary date T_0 , with $T_0 < T_1 < \dots < T_n$, as the start date and valued our cash flows until maturity. Now we can firstly value our swaps, and we can solve for a forward swap rate.

Let's firstly tackle our forward swap rate, which we will define as being $y_t[T_0, T_n]$. The forward swap rate is the fixed rate that makes a value of a swap from T_0 to T_n worth zero at time t . where

$$V_K^{FXD}(t) = V_K^{FL}(t) \rightarrow y_t[T_0, T_n] = \frac{Z(t, T_0) - Z(t, T_n)}{P_t[T, T + \alpha]}$$

We can also right the most right hand side of the equality as

$$\frac{\sum_{i=1}^n \alpha S_t[T_{i-1}, T_i] Z(t, T_i)}{\sum_{i=1}^n \alpha Z(t, T_i)}$$

This is a very interesting result as it means we can write our forward swap rate, $y_t[T_0, T_n]$ as the weighted average of forward rates.

Now we can value our swaps for any start date T_0 . As we have established above the value is the difference between the two legs of the swap.

$$V_k^{SW}(t) = Z(0, T) - Z(t, T_n) - KP_t[T, T_n]$$

We can use our previous result to write our value of the swap as

$$V_k^{SW}(t) = y_t[T_0, T_n]P_t[T, T_n] - KP_t[T, T_n] = (y_t[T_0, T_n] - K)P_t[T, T_n]$$

Here is quite a nice symmetry between a forward contract's value at time t : $V_K(t, T) = (F(t, T) - K)e^{-r(T-t)}$, where $F(t, T)$ is the forward price of the asset.

The par swap rate is when $t = T_0$. This is the fixed rate at which one can invest 1 at the time T_0 until maturity, T_n receiving fixed payments, then paying the 1 pound back at T_n .

3 Conclusion

To conclude, in this document I highlight the method I learnt from "An Introduction to Quantitative Finance" by Stephen Blyth and some additional reading from Howard Cobb's book on interest rate derivatives to value interest rate swaps and Forward Rate Agreements. In the future I hope to add the methods behind pricing Basis and Cross-Currency swaps to this document as I continue to work on this project.